PROTOFIRE

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Smart contract audit
rain.metadata

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Project description

This repository, rain.metadata, is a native Solidity library and registry system that implements the Rain Metadata V1 specification. Its primary purpose is to provide a standard, gas-efficient mechanism for associating off-chain data (such as documentation, ABIs, and source code) with on-chain artifacts.

The core component is the MetaBoard, a permissionless contract that allows any account to emit metadata events. The repository also includes LibMeta and IDescribedByMetaV1 to facilitate metadata validation, hashing, and the creation of "self-describing" contracts. This architecture enables a verifiable link between smart contracts and their metadata without incurring the high cost of on-chain storage.

Executive summary

Type	Library
Languages	Solidity
Methods	Architecture Review, Manual Review, Unit Testing, Functional Testing, Automated Review
Documentation	README.md
Repositories	https://github.com/rainlanguage/rain.metadata

Reviews

Date	Repository	Commit
02/02/26	rain.metadata	fde2333cdf92d0879164be4eacb1eb844cea79d9

Scope

Contracts
src/concrete/MetaBoard.sol
src/interface/deprecated/IMetaBoardV1.sol
src/interface/deprecated/IMetaV1.sol

src/interface/unstable/IMetaBoardV1_2.sol
src/interface/unstable/IMetaV1_2.sol
src/interface/IDescribedByMetaV1.sol
src/lib/LibDescribedByMeta.sol
src/lib/LibMeta.sol

Technical analysis and findings

Critical	0
High	0
Medium	0
Low	0
Informational	0

Security findings

**** Critical

No critical severity issue found.

*** High

No high severity issue found.

** Medium

No medium severity issue found.

* Low

No low severity issue found.

Informational

No informational severity issue found.

General Risks

- Permissionless Registry / Spam: The MetaBoard contract is fully permissionless, meaning any address can emit metadata for any subject. There is a risk of "registry poisoning" where malicious actors emit spam or misleading metadata. Off-chain indexers and tooling must implement robust filtering (e.g., verifying that the sender is authorized for the subject) to ignore untrusted data.

- Data Availability: The system enforces the integrity of the metadata (via hashing) but not its availability. Since the full metadata payload is emitted in event logs or stored off-chain, there is a risk that the underlying data becomes unavailable (e.g., if history is not preserved or off-chain storage fails), leaving only an opaque hash on-chain.

Approach and methodology

To establish a uniform evaluation, we define the following terminology in accordance with the OWASP Risk Rating
Methodology:

	Likelihood Indicates the probability of a specific vulnerability being discovered and exploited in real-world scenarios
	Impact Measures the technical loss and business repercussions resulting from a successful attack
	Severity Reflects the comprehensive magnitude of the risk, combining both the probability of occurrence (likelihood) and the extent of potential consequences (impact)

Likelihood and impact are divided into three levels: High H, Medium M, and Low L. The severity of a risk is a blend of these two factors, leading to its classification into one of four tiers: Critical, High, Medium, or Low.

When we identify an issue, our approach may include deploying contracts on our private testnet for validation through testing. Where necessary, we might also create a Proof of Concept PoC to demonstrate potential exploitability. In particular, we perform the audit according to the following procedure:

	Advanced DeFi Scrutiny We further review business logics, examine system operations, and place DeFi-related aspects under scrutiny to uncover possible pitfalls and/or bugs
	Semantic Consistency Checks We then manually check the logic of implemented smart contracts and compare with the description in the white paper.
	Security Analysis The process begins with a comprehensive examination of the system to gain a deep understanding of its internal mechanisms, identifying any irregularities and potential weak spots.